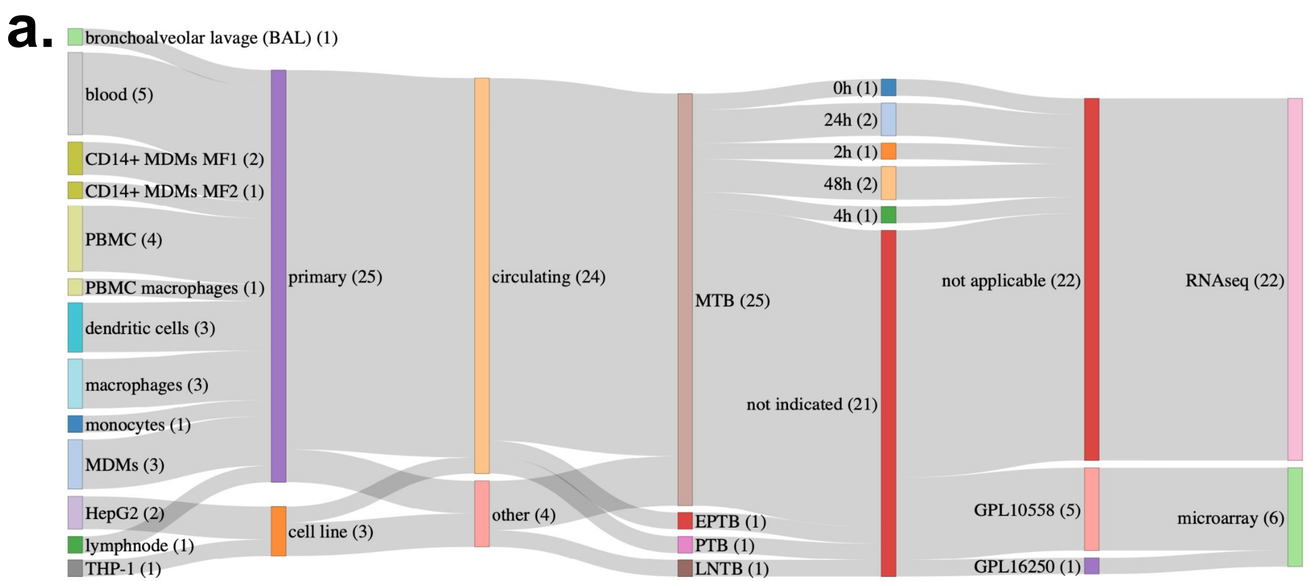
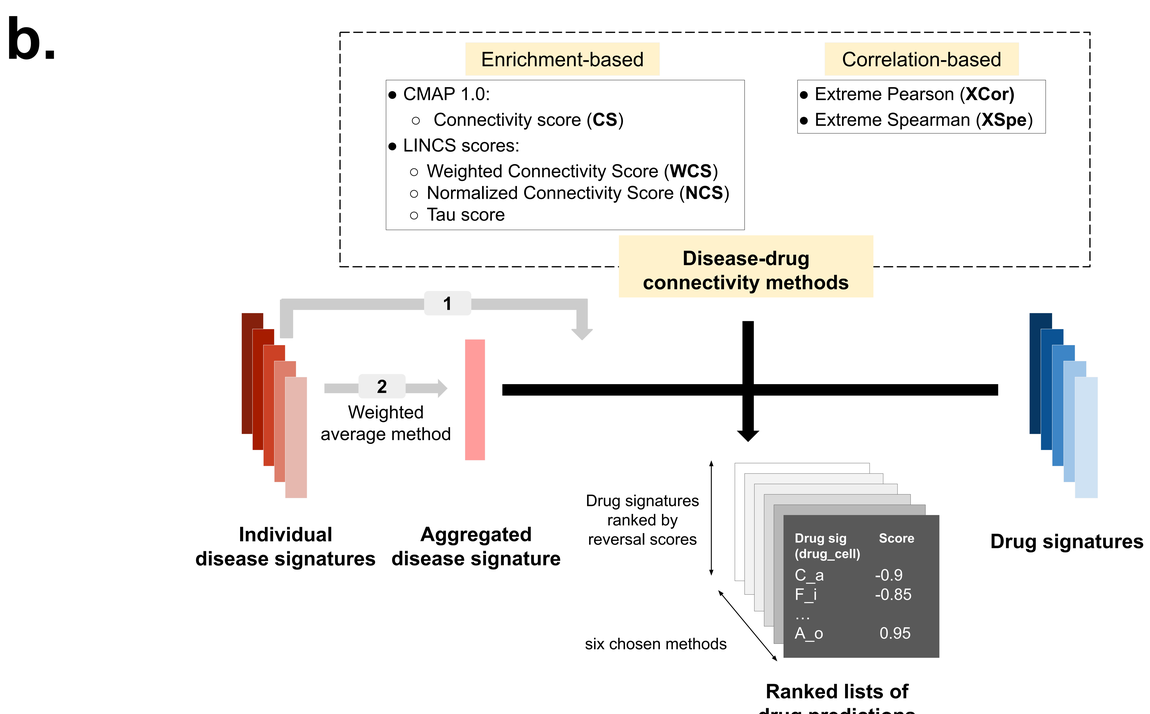
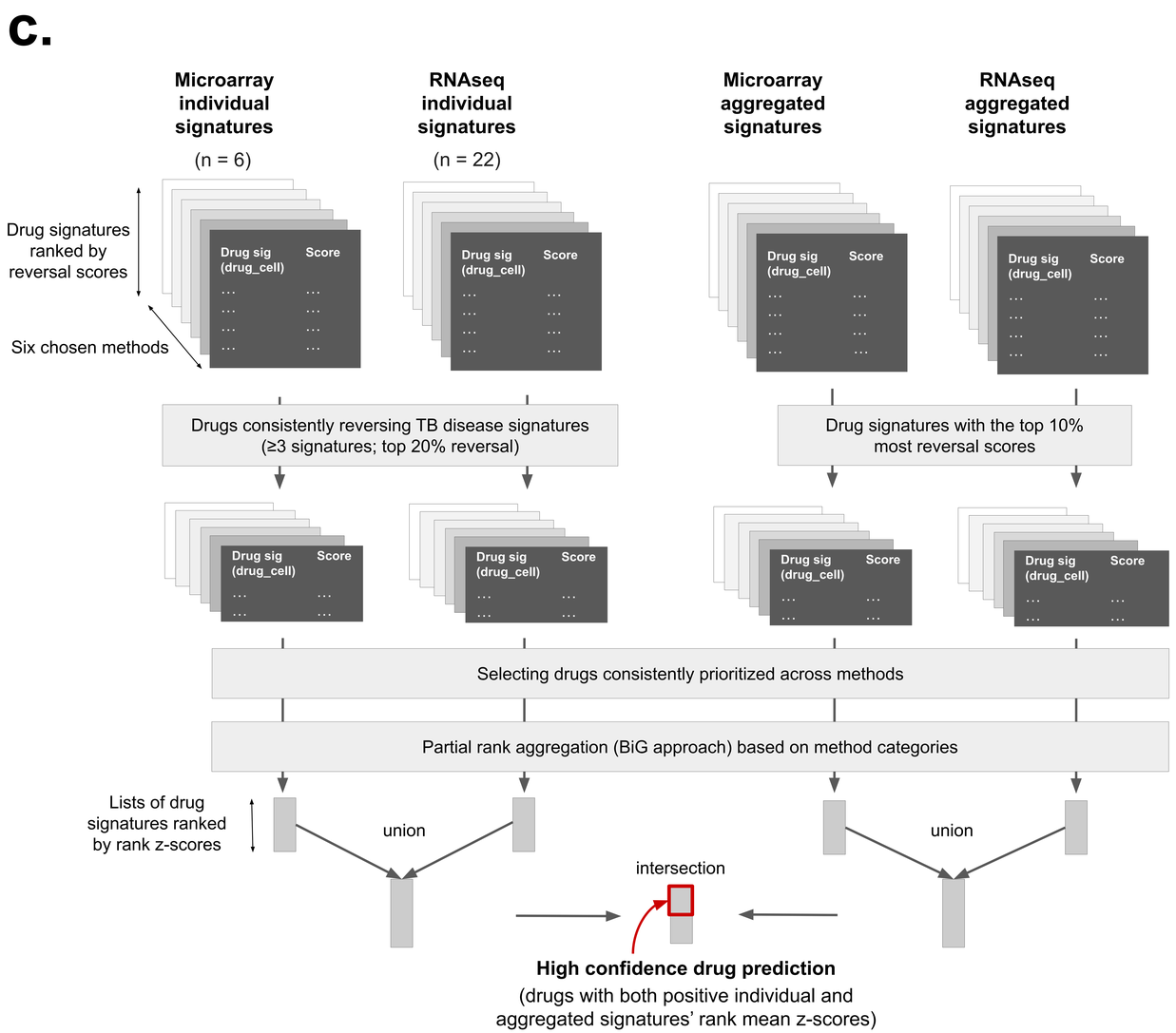

a.
bronchoalveolar lavage (BAL) (1)
blood (5)
CD14+ MDMs MF1 (2)
CD14+ MDMs MF2 (1)
PBMC (4)
PBMC macrophages (1)
dendritic cells (3)
macrophages (3)
monocytes (1)
MDMs (3)
HepG2 (2)
lymphnode (1)
THP-1 (1)
primary (25)
cell line (3)
circulating (24)
other (4)
MTB (25)
not indicated (21)
EPTB (1)
PTB (1)
LNTB (1)
0h (1)
24h (2)
2h (1)
48h (2)
4h (1)
not applicable (22)
GPL10558 (5)
GPL16250 (1)
RNAseq (22)
microarray (6)


b.
Enrichment-based
CMAP 1.0:
Connectivity score (CS)
LINCS scores:
Weighted Connectivity Score (WCS)
Normalized Connectivity Score (NCS)
Tau score
Correlation-based
Extreme Pearson (XCor)
Extreme Spearman (XSpe)
Disease-drug connectivity methods
1
2
Weighted average method
Individual disease signatures
Aggregated disease signature
Drug signatures ranked by reversal scores
six chosen methods
Drug sig (drug_cell)
Score
C_a -0.9
F_i -0.85
A_o 0.95
Ranked lists of drug predictions
Drug signatures


c.
Microarray individual signatures
(n = 6)
RNAseq individual signatures
(n = 22)
Microarray aggregated signatures
RNAseq aggregated signatures
Drug signatures ranked by reversal scores
Six chosen methods
Drug sig (drug_cell)
Score
Drugs consistently reversing TB disease signatures (≥3 signatures; top 20% reversal)
Drug signatures with the top 10% most reversal scores
Selecting drugs consistently prioritized across methods
Partial rank aggregation (BiG approach) based on method categories
Lists of drug signatures ranked by rank z-scores
union
intersection
High confidence drug prediction
(drugs with both positive individual and aggregated signatures' rank mean z-scores)